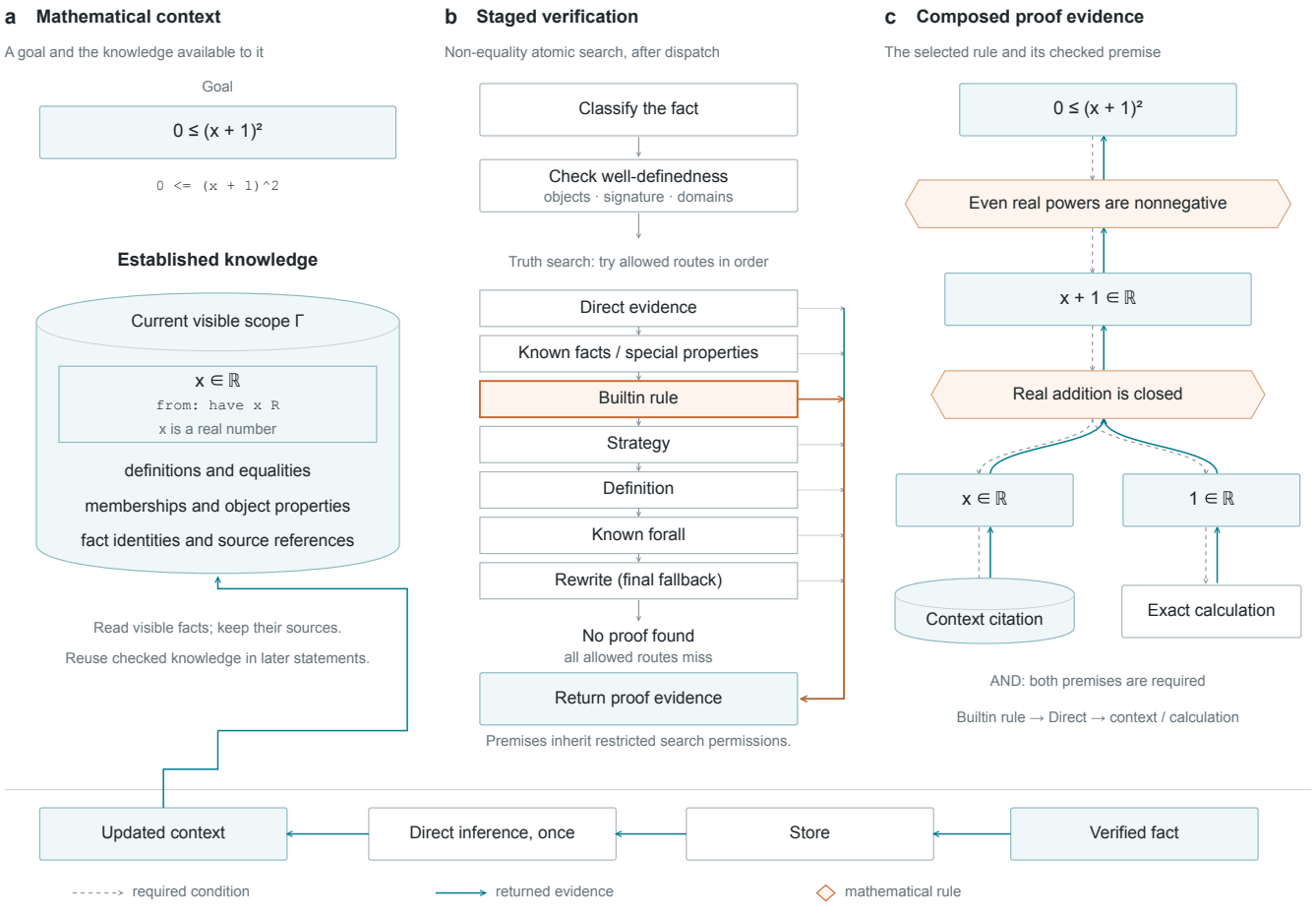

a
Mathematical context
A goal and the knowledge available to it
Goal
0 ≤ (x + 1)²
0 <= (x + 1)^2
Established knowledge
Current visible scope Γ
x ∈ ℝ
from: have x R
x is a real number
definitions and equalities
memberships and object properties
fact identities and source references
Read visible facts; keep their sources.
Reuse checked knowledge in later statements.
b
Staged verification
Non-equality atomic search, after dispatch
Classify the fact
Check well-definedness
objects · signature · domains
Truth search: try allowed routes in order
Direct evidence
Known facts / special properties
Builtin rule
Strategy
Definition
Known forall
Rewrite (final fallback)
No proof found
all allowed routes miss
Return proof evidence
Premises inherit restricted search permissions.
c
Composed proof evidence
The selected rule and its checked premise
0 ≤ (x + 1)²
Even real powers are nonnegative
x + 1 ∈ ℝ
Real addition is closed
x ∈ ℝ
1 ∈ ℝ
Context citation
Exact calculation
AND: both premises are required
Builtin rule → Direct → context / calculation
Updated context
Direct inference, once
Store
Verified fact
required condition
returned evidence
mathematical rule